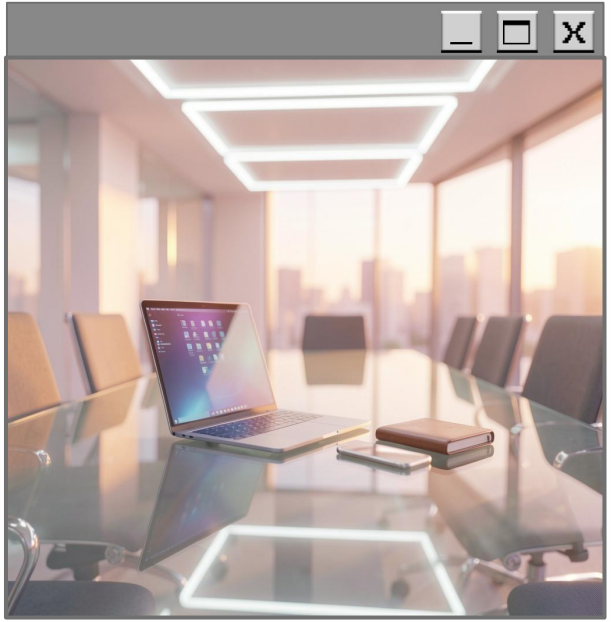
Aisha Nomani, An Chu, Aurelia Vo, Sai Han Shan Pha, & Mohammed Aljumai

PrecisionParts Manufacturing



Supply Chain & Quality Analytics

Team: The Leftside Undergrads

The Business Problem

+15%

Defect Rate Increase

$28M annual cost

$30M

Inventory Carrying

25% above benchmark

200

Suppliers Zero Visibility

Manual quarterly scores

180h

Unplanned Downtime /Factory

$50k per hour

Data Lake Architecture Diagram

Batch Pipeline

Course Repository Data → Local data/raw/ → HDFS Landing Zone → Spark Batch Processing → HDFS Curated Zone → Spark Joins and Aggregations → HDFS Analytics Zone → Reports, Slides, and Business Insights

Batch + Steaming share the same
HDFS + Parquet foundation

Streaming Pipeline

Equipment Sensor Data → Kafka Topic → Spark Structured Streaming → Real-Time Equipment Alerts → Airflow Streaming Monitor

Why This Architecture?

- Designed to solve real business problems first, not just use tools
- Supports both batch analytics + real-time monitoring in one system

Handles Two Workloads

Batch (Spark): historical analysis, joins, reporting

Streaming (Kafka + Spark Streaming): real-time machine alerts

Solves Data Silos Problem

Connects production, quality, supplier, inventory, and sensor data

Enables root-cause analysis across systems

Built for Scale

HDFS → *distributed storage*

Spark → *large-scale processing*

Kafka → *high-throughput streaming*

Parquet → *fast analytics queries*

Enables Real-Time + Historical Insights

Batch layer → *deep business insights*

Streaming layer → *instant anomaly detection*

Reliable & Maintainable

Airflow orchestrates full pipeline

Retry + idempotent design ensures stability

Stage 1 - Data Lake Foundation

Landing Zone:

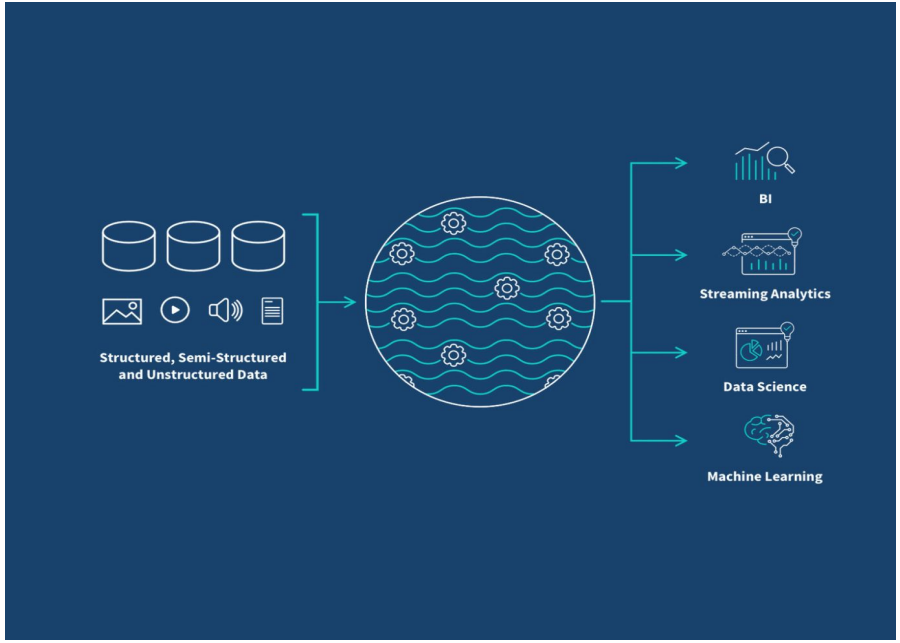
- Raw CSV.GZ / JSON.GZ files
- No transformation

Curated Zone:

- Cleaned using Spark
- Parquet format
- Deduplication + validation

Analytics Zone:

- Final aggregated datasets
- Used for reporting + insights



Stage 2 - Spark Batch Transformation

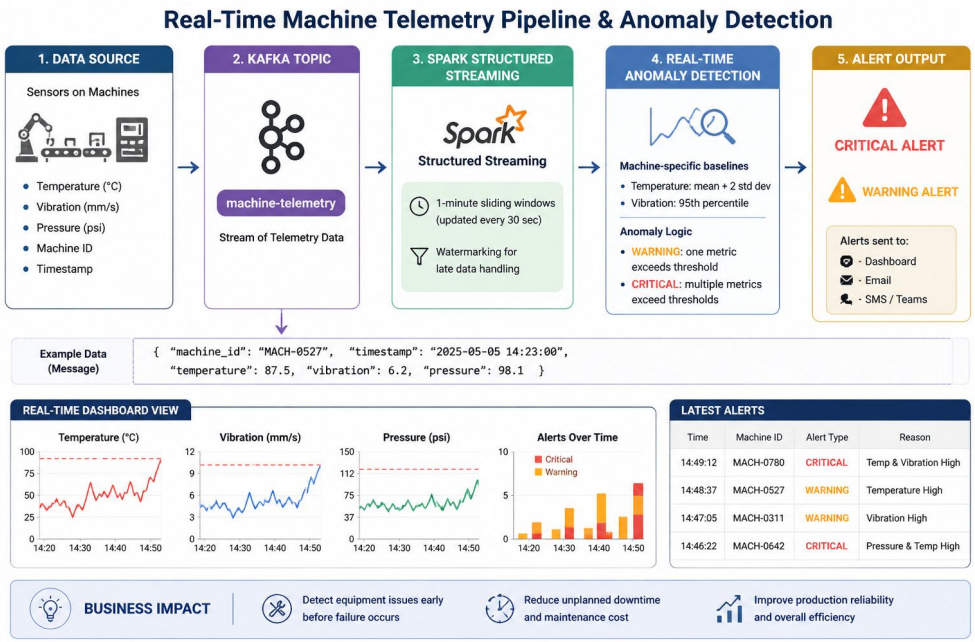
Data Cleaning:

- Remove duplicates
- Handle missing values
- Type casting
- Sensor validation

Output:

- Clean curated datasets ready for analytics

Stage 3 - Real-Time Streaming Pipeline



Kafka Topic:

- `machine-telemetry`

Data:

- Temperature
- Vibration
- Pressure
- Machine ID + timestamp

Processing:

- Spark Structured Streaming
- 1-minute sliding windows
- Real-time anomaly detection

Stage 4 - Exploration

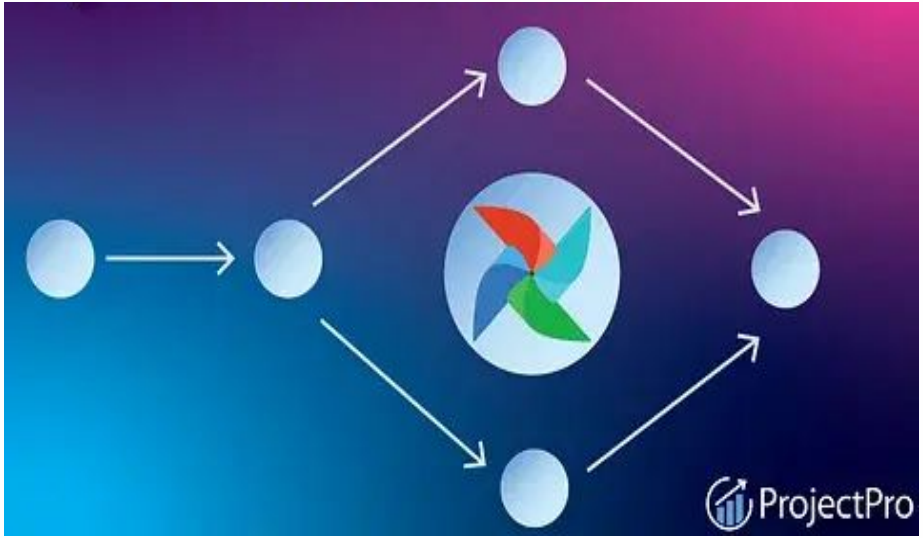
- *Orchestrates end-to-end pipeline execution using Apache Airflow*
- *Manages batch DAG (Spark jobs) + streaming health monitoring*

Batch Pipeline:

- Validate data availability
- Run Spark cleaning + transformations
- Execute analytics jobs
- Generate final reports

Streaming Monitoring:

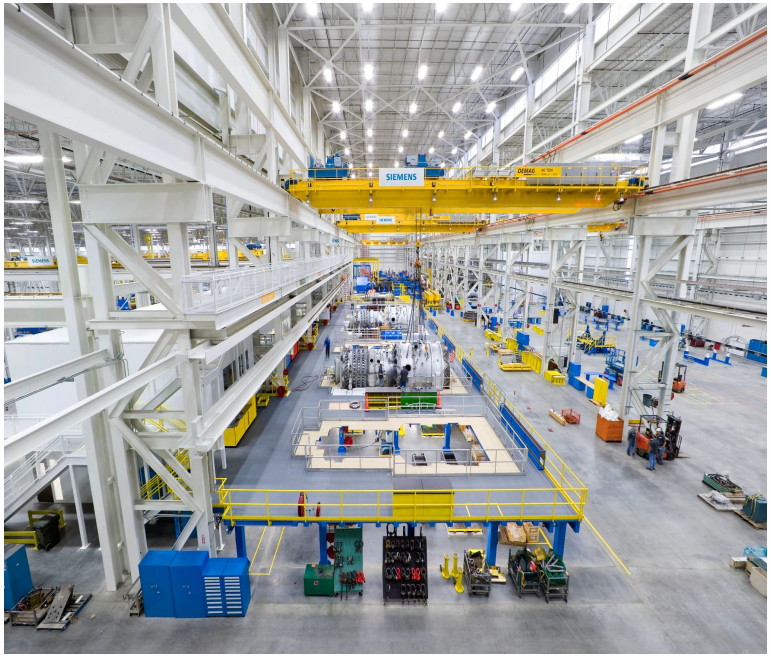
- Checks streaming output every 5 minutes
- Detects missing or delayed sensor data



Key Role:

Ensures the entire pipeline runs reliably, in correct order, with failure recovery and monitoring.

Business Insights - Defect Analysis



- The pipeline computed defect rates across factories, production lines, and shifts by linking production and quality inspection data
- Results show that defect rates are not uniformly distributed across operations
- Night shifts consistently show higher defect rates across multiple factories
- Certain combinations of factory, production line, and shift contribute most to defect concentration

=> Defect patterns are concentrated in specific operational conditions rather than spread evenly across the system

Impact:

- Provides a data-backed basis for targeted interventions such as staffing adjustments, training programs, and shift scheduling changes instead of broad company-wide actions

Business Insights - Suppliers & Inventory

Defects:

- Night shifts have higher defect rates
- Few factories contribute most defects

Suppliers:

- Supplier scores are lagging indicators
- High-risk suppliers identified

Inventory:

- Overstock + stockout imbalance
- High stockout cost risk

Machines:

- Sensor anomalies predict downtime + defects

Predictive Maintenance & Anomaly Detection

- *Uses real-time sensor data from Kafka* (machine-telemetry)
- *Processes data with Spark Structured Streaming (1-min windows)*
- *Applies machine-specific baselines (mean + std dev, 95th percentile)*

Anomaly Detection:

- **WARNING** → single metric exceeds threshold
- **CRITICAL** → multiple metrics exceed thresholds

Impact:

- Enables real-time detection of equipment issues, reducing unplanned downtime and maintenance costs.

Lessons Learned

What Worked:

- Lambda architecture fit perfectly
- Spark handled large-scale processing well
- Kafka enabled real-time pipeline

What Was Hard:

- Debugging distributed systems
- Streaming vs batch consistency
- Managing state in real-time processing

What We Would Improve:

- Move to AWS (S3, EMR, MSK, MWAA)
- Add ML-based prediction models
- Build BI dashboards (Power BI / Tableau)